

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37



Department of Computer Engineering

Class and Division	CS-A
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Assignment Number	2
Assignment Title	Design and execute SQL queries on a Hospital Management System database. The database should consist of the following: Patient Table (ID, Name, DOB, Gender, Phone, Address). Doctor Table (ID, Name, Specialization, Phone). Appointment Table (ID, PatientID, DoctorID, Date, Status).

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Theory:

The following SQL concepts are used:

- **Conditional Operators:** =, >, <, >=, <=
- **Logical Operators:** AND, OR, NOT
- **Pattern Matching:** LIKE, NOT LIKE
- **Set Operators:** IN, NOT IN
- **Range Operator:** BETWEEN . . . AND
- **Null Checks:** IS NULL, IS NOT NULL
- **Sorting:** ORDER BY
- **Grouping:** GROUP BY
- **Filtering Groups:** HAVING
- **Aggregate Functions:** COUNT(), SUM(), AVG(), MAX(), MIN()
- **Single Row Functions:** Date functions and String functions.

These concepts help retrieve, filter, sort, and analyze hospital data efficiently.

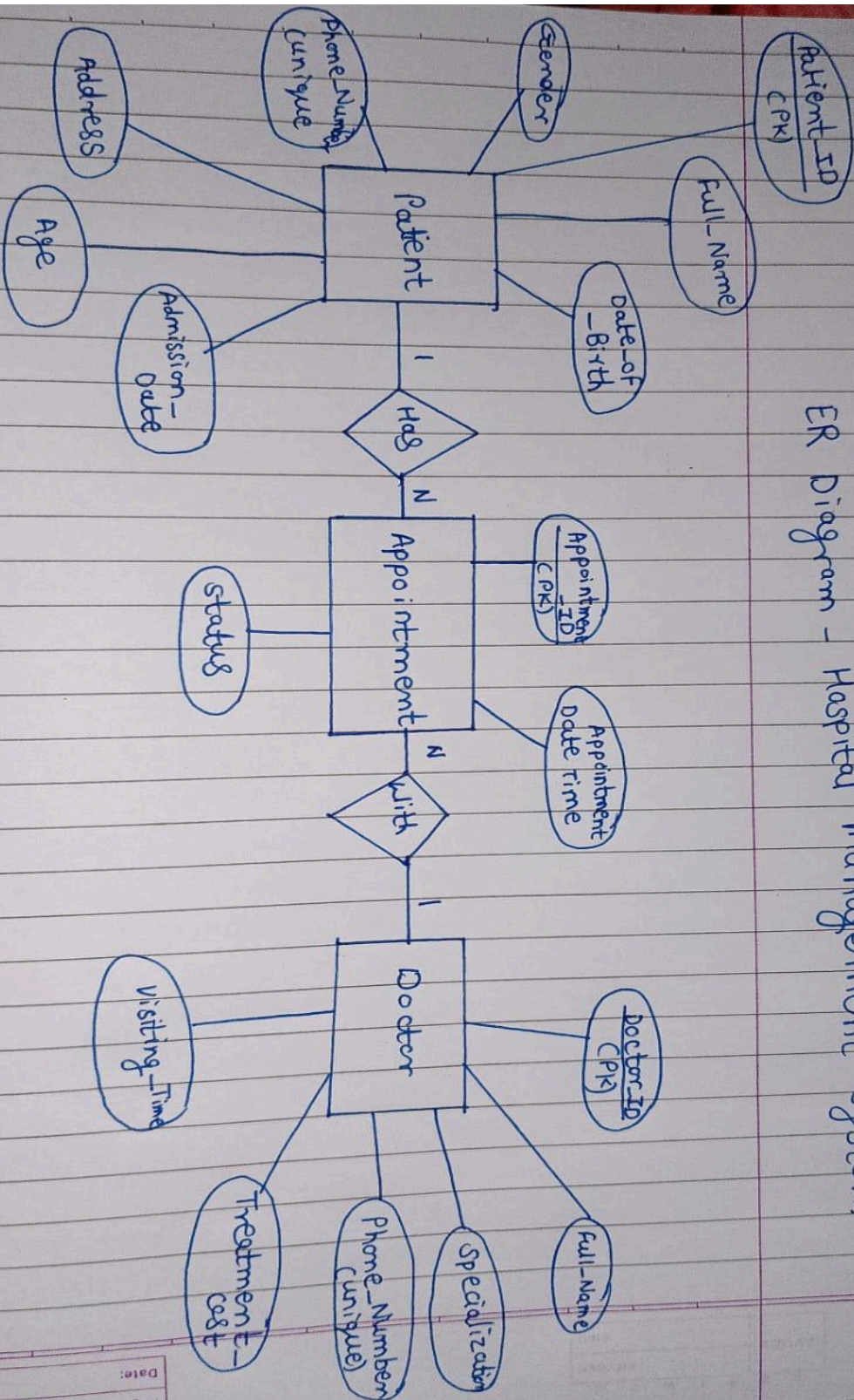
Entities

1. Patient
2. Doctor
3. Appointment

Relationships

- One Patient can have many Appointments.
- One Doctor can have many Appointments.
- Appointment links Patient and Doctor.

ER Diagram - Hospital Management System



Input:

Relation Details

Patient Table

DESC patient;

Doctor Table

DESC doctor;

Appointment Table

DESC appointment;

SQL Queries Used

1. Display male patients.
2. Display appointments on or after a given date.
3. Display female patients from Pune with phone number.
4. Display patients from Mumbai or Pune.
5. Display doctors who are not cardiologists.
6. Display patients whose names start with A.
7. Display doctors with specialization containing Neuro.
8. Display patients whose address does not contain Maharashtra.
9. Display patients from Pune.
10. Display doctors not specialized in Cardiology or Dermatology.
11. Display patients whose DOB lies between two dates.
12. Display patients with NULL phone numbers.

13. Display doctors whose phone numbers are available.
14. Display doctors ordered by name.
15. Display doctors having more than five appointments.
16. Display minimum and maximum appointment dates.

Output:

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 9

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE hospital_assignment;

Database changed

mysql> select * from patient where gender="male";

```
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
| Patient_ID | Full_Name   | Date_of_Birth | Gender | Phone_Number | Address | Age |
| Admission_Date |
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
|      1 | Rahul Sharma | 2001-05-15   | Male   | 9999999999   | Pune    | 25 |
2026-07-01 |
|      3 | Amit Kumar   | 1979-02-10   | Male   | 9876512345   | Kolhapur | 47 |
2026-07-05 |
|      5 | Rohan Desai  | 1987-07-18   | Male   | 9876534567   | Sangli   | 39 |
2026-07-10 |
|      7 | Vikram Singh | 1974-09-14   | Male   | 9876556789   | Nagpur   | 51 |
2026-07-15 |
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
4 rows in set (0.08 sec)
```

mysql> select * from appointment where Appointment_DateTime >= '2025-01-01';

```
+-----+-----+-----+-----+-----+
| Appointment_ID | Patient_ID | Doctor_ID | Appointment_DateTime | Status |
+-----+-----+-----+-----+-----+
|      1 |      1 |      1 | 2026-08-02 09:00:00 | Completed |
```

2	2	2	2026-08-02 10:00:00	Completed
3	3	3	2026-08-03 11:00:00	Scheduled
4	4	4	2026-08-03 12:00:00	Cancelled
5	5	5	2026-08-04 09:30:00	Scheduled
6	6	6	2026-08-04 10:30:00	Completed
7	7	7	2026-08-05 11:30:00	Scheduled
8	8	8	2026-08-05 12:30:00	Cancelled
10	10	2	2026-08-06 10:15:00	Scheduled

9 rows in set (0.01 sec)

```
mysql> select * from patient where Gender='female' and Address='pune' and
Phone_Number IS NOT NULL;
Empty set (0.02 sec)
```

```
mysql> select * from patient where Address='mumbai' OR Address='pune';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03

2 rows in set (0.00 sec)

```
mysql> select * from doctor where Specialization IS NOT cardiologist;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual
that corresponds to your MySQL server version for the right syntax to use near
'cardiologist' at line 1
```

```
mysql> SELECT * FROM doctor
-> WHERE NOT Specialization='Cardiology';
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
-----------	-----------	----------------	--------------	----------------	---------------

2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
4	Dr. Joshi	Dermatology	9876600004	2500.00	10:30:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

6 rows in set (0.01 sec)

```
mysql> select * from patient where Full_Name like 'A%';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12

2 rows in set (0.00 sec)

```
mysql> select * from doctor WHERE Specialization LIKE '%Neuro%';
```

3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
---	-----------	-----------	------------	---------	----------

1 row in set (0.00 sec)

```
mysql> SELECT * FROM patient
-> WHERE Address NOT LIKE '%Maharashtra%';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
------------	-----------	---------------	--------	--------------	---------	-----	----------------

```

+-----+-----+-----+-----+-----+-----+-----+
-----+
|      1 | Rahul Sharma | 2001-05-15 | Male | 9999999999 | Pune | 25 |
2026-07-01 |
|      2 | Priya Patil   | 1994-08-22 | Female | 9876501234 | Mumbai | 31 |
2026-07-03 |
|      3 | Amit Kumar    | 1979-02-10 | Male | 9876512345 | Kolhapur | 47 |
2026-07-05 |
|      4 | Sneha Joshi   | 1996-11-05 | Female | 9876523456 | Satara | 29 |
2026-07-08 |
|      5 | Rohan Desai   | 1987-07-18 | Male | 9876534567 | Sangli | 39 |
2026-07-10 |
|      6 | Anjali Kulkarni | 1998-03-25 | Female | 8888888888 | Nagpur | 28 |
2026-07-12 |
|      7 | Vikram Singh  | 1974-09-14 | Male | 9876556789 | Nagpur | 51 |
2026-07-15 |
|      8 | Neha Verma    | 1991-12-30 | Female | 9876567890 | Solapur | 34 |
2026-07-18 |
|     10 | Pooja More    | 1993-10-12 | Female | 9876589012 | Ahmednagar |
32 | 2026-07-22 |
+-----+-----+-----+-----+-----+-----+-----+
-----+
9 rows in set (0.00 sec)

```

```
mysql> select * from patient where Address='Pune';
```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+
| Patient_ID | Full_Name | Date_of_Birth | Gender | Phone_Number | Address |
Age | Admission_Date |
+-----+-----+-----+-----+-----+-----+-----+
-----+
|      1 | Rahul Sharma | 2001-05-15 | Male | 9999999999 | Pune | 25 |
2026-07-01 |
+-----+-----+-----+-----+-----+-----+-----+
-----+
1 row in set (0.00 sec)

```

```
mysql> SELECT * FROM doctor
```

```
-> WHERE Specialization NOT IN ('Cardiology','Dermatology');
```

```

+-----+-----+-----+-----+-----+-----+-----+

```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

5 rows in set (0.00 sec)

```
mysql> select * from patient where Date_of_Birth BETWEEN 1980-01-01 AND 2000-12-31;
Empty set, 2 warnings (0.01 sec)
```

```
mysql> SELECT * FROM patient
-> WHERE Date_of_Birth
-> BETWEEN '1980-01-01' AND '2000-12-31';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22

6 rows in set (0.01 sec)

```
mysql> select * from patient where Phone_Number is NULL;
Empty set (0.02 sec)
```

```
mysql> select * from doctor where Phone_Number is not NULL;
```

```
+-----+-----+-----+-----+-----+-----+
| Doctor_ID | Full_Name   | Specialization | Phone_Number | Treatment_Cost | Visiting_Time |
+-----+-----+-----+-----+-----+-----+
| 1 | Dr. Sharma | Cardiology     | 9876600001   | 5000.00 | 09:00:00 |
| 2 | Dr. Mehta  | Orthopedic     | 9876600002   | 3500.00 | 09:30:00 |
| 3 | Dr. Patil  | Neurology      | 9876600003   | 6500.00 | 10:00:00 |
| 4 | Dr. Joshi  | Dermatology    | 9876600004   | 2500.00 | 10:30:00 |
| 5 | Dr. Kulkarni | Cardiology     | 9876600005   | 5500.00 | 11:00:00 |
| 6 | Dr. Deshmukh | Pediatrics     | 9876600006   | 3000.00 | 11:30:00 |
| 7 | Dr. Shah   | ENT            | 9876600007   | 2900.00 | 12:00:00 |
| 8 | Dr. Reddy  | General Medicine | 9876600008   | 2000.00 | 12:30:00 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> select * from doctor order by Full_Name DESC;
```

```
+-----+-----+-----+-----+-----+-----+
| Doctor_ID | Full_Name   | Specialization | Phone_Number | Treatment_Cost | Visiting_Time |
+-----+-----+-----+-----+-----+-----+
| 1 | Dr. Sharma | Cardiology     | 9876600001   | 5000.00 | 09:00:00 |
| 7 | Dr. Shah   | ENT            | 9876600007   | 2900.00 | 12:00:00 |
| 8 | Dr. Reddy  | General Medicine | 9876600008   | 2000.00 | 12:30:00 |
+-----+-----+-----+-----+-----+-----+
| 3 | Dr. Patil  | Neurology      | 9876600003   | 6500.00 | 10:00:00 |
| 2 | Dr. Mehta  | Orthopedic     | 9876600002   | 3500.00 | 09:30:00 |
| 5 | Dr. Kulkarni | Cardiology     | 9876600005   | 5500.00 | 11:00:00 |
| 4 | Dr. Joshi  | Dermatology    | 9876600004   | 2500.00 | 10:30:00 |
| 6 | Dr. Deshmukh | Pediatrics     | 9876600006   | 3000.00 | 11:30:00 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.02 sec)
```

```
mysql> SELECT Doctor_ID, COUNT(*) AS Total_Appointments
-> FROM appointment
```

```
-> GROUP BY Doctor_ID
-> HAVING COUNT(*) > 5;
Empty set (0.02 sec)
```

```
mysql> SELECT
-> MIN(Appointment_DateTime) AS Earliest_Appointment,
-> MAX(Appointment_DateTime) AS Latest_Appointment
-> FROM appointment;
```

```
+-----+-----+
| Earliest_Appointment | Latest_Appointment |
+-----+-----+
| 2026-08-02 09:00:00 | 2026-08-06 10:15:00 |
+-----+-----+
1 row in set (0.01 sec)
```

```
mysql>
```

Q1. Write a query to display all male patients.

```
mysql> USE hospital_assignment;
Database changed
mysql> select * from patient where gender="male";
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15

```
4 rows in set (0.08 sec)
```

Q2.Display appointments scheduled on or after 2025-01-01.

```
mysql> select * from appointment where Appointment_DateTime >= '2025-01-01';
```

Appointment_ID	Patient_ID	Doctor_ID	Appointment_DateTime	Status
1	1	1	2026-08-02 09:00:00	Completed
2	2	2	2026-08-02 10:00:00	Completed
3	3	3	2026-08-03 11:00:00	Scheduled
4	4	4	2026-08-03 12:00:00	Cancelled
5	5	5	2026-08-04 09:30:00	Scheduled
6	6	6	2026-08-04 10:30:00	Completed
7	7	7	2026-08-05 11:30:00	Scheduled
8	8	8	2026-08-05 12:30:00	Cancelled
10	10	2	2026-08-06 10:15:00	Scheduled

```
9 rows in set (0.01 sec)
```

Q3.Display female patients who live in Pune AND have a phone number.

```
mysql> select * from patient where Gender='female' and Address='pune' and Phone_Number IS NOT NULL;
Empty set (0.02 sec)
```

Q4.Display patients who live in Mumbai OR Pune.

```
mysql> select * from patient where Address='mumbai' OR Address='pune';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03

```
2 rows in set (0.00 sec)
```

Q5.List all doctors who are NOT cardiologists.

```
mysql> select * from doctor where Specialization IS NOT cardiologist;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'c
ardiologist' at line 1
mysql> SELECT * FROM doctor
-> WHERE NOT Specialization='Cardiology';
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
4	Dr. Joshi	Dermatology	9876600004	2500.00	10:30:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

```
6 rows in set (0.01 sec)
```

Q6.Display patients whose name starts with 'A'.

```
mysql> select * from patient where Full_Name like 'A%';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12

```
2 rows in set (0.00 sec)
```

Q7.Find doctors whose specialization contains the word 'Neuro'.

```
mysql> select * from doctor WHERE Specialization LIKE '%Neuro%';
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00

```
1 row in set (0.00 sec)
```

Q8.Display patients whose address does not contain the word 'Maharashtra'.

```
mysql> SELECT * FROM patient
-> WHERE Address NOT LIKE '%Maharashtra%';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22

```
9 rows in set (0.00 sec)
```

Q9.Display patients who are from cities (Pune).

```
mysql> select * from patient where Address='Pune';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01

```
1 row in set (0.00 sec)
```

Q10.Display doctors who are not specialized in ('Cardiologist', 'Dermatologist').

```
mysql> SELECT * FROM doctor
-> WHERE Specialization NOT IN ('Cardiology','Dermatology');
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

```
5 rows in set (0.00 sec)
```

Q11.Display patients whose DOB is between 1980-01-01 and 2000-12-31.

```
mysql> SELECT * FROM patient
-> WHERE Date_of_Birth
-> BETWEEN '1980-01-01' AND '2000-12-31';
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22

```
6 rows in set (0.01 sec)
```

Q12.Display patients who have not provided their phone number.

```
mysql> select * from patient where Phone_Number is NULL;
Empty set (0.02 sec)
```

Q13.List all doctors whose phone number is available (not null).

```
mysql> select * from doctor where Phone_Number is not NULL;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
1	Dr. Sharma	Cardiology	9876600001	5000.00	09:00:00
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
4	Dr. Joshi	Dermatology	9876600004	2500.00	10:30:00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	11:00:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

```
8 rows in set (0.00 sec)
```

Q14.Display doctors ordered alphabetically by name in descending order.

```
mysql> select * from doctor order by Full_Name DESC;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
1	Dr. Sharma	Cardiology	9876600001	5000.00	09:00:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00
3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	11:00:00
4	Dr. Joshi	Dermatology	9876600004	2500.00	10:30:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00

```
8 rows in set (0.02 sec)
```

Q15.Display doctors who have more than 5 appointments (use HAVING).

```
mysql> SELECT Doctor_ID, COUNT(*) AS Total_Appointments
-> FROM appointment
-> GROUP BY Doctor_ID
-> HAVING COUNT(*) > 5;
Empty set (0.02 sec)
```

Q16.Find the earliest (MIN) and latest (MAX) appointment dates.

```
mysql> SELECT
-> MIN(Appointment_DateTime) AS Earliest_Appointment,
-> MAX(Appointment_DateTime) AS Latest_Appointment
-> FROM appointment;
+-----+
| Earliest_Appointment | Latest_Appointment |
+-----+
| 2026-08-02 09:00:00   | 2026-08-06 10:15:00 |
+-----+
1 row in set (0.01 sec)

mysql> |
```

Learning Outcome

After completing this assignment, the following concepts were understood:

- Using conditional operators in SQL.
- Applying logical operators (AND, OR, NOT).
- Searching records using LIKE and NOT LIKE.
- Using IN and NOT IN for multiple values.
- Retrieving data within a range using BETWEEN.
- Checking NULL values using IS NULL and IS NOT NULL.
- Sorting records using ORDER BY.
- Using aggregate functions such as COUNT(), MIN(), and MAX().
- Grouping data using GROUP BY and filtering grouped data using HAVING.

Conclusion

The Hospital Management System database was successfully queried using different SQL operators and functions. Various SQL statements were executed to retrieve records based on conditions, sort data, search patterns, perform aggregation, and analyze information stored in the Patient, Doctor, and Appointment tables. This assignment improved the understanding of SQL query writing, conditional filtering, and data retrieval techniques in a relational database system.